ORIGINAL RESEARCH ARTICLE

High-Throughput Experiment and Numerical Simulation to Study Solidification Cracking in 2195 Aluminum Alloy Welds

M. AGILAN, K. SATYAMSHRESHTA, D. SIVAKUMAR, and G. PHANIKUMAR

Solidification cracking during welding is a severe concern in aluminum alloys. Hot-cracking criteria have been successful in grading the crack susceptibility of an alloy. However, they do not offer a safe window of processing conditions for a crack-free weld. This work intends to determine the processing window for crack-free welding using a high-throughput experiment coupled with computer simulation. An experimental setup to produce a spiral welding path was developed to obtain a wide variation in heat input resulting in a range of cooling rates ($\dot{T}$) and temperature gradients (G) in a single experiment. Simulations using OpenFOAM® were performed to determine the spatial variation in $\dot{T}$ and G. A parametric space constructed using cooling rate and thermal gradient extracted from various points along the spiral path was used to map the regime for a crack-free weld. The results were validated using linear welds. The results were rationalized using microstructure evolution, and micro-segregation predicted using phase-field simulations.

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I. INTRODUCTION

HOT cracking during fusion welding is a major concern for several alloys of industrial importance. These alloys include age-hardenable aluminum alloys, stainless steel, and superalloys. Cracking occurs towards the last stage of solidification predominantly due to thermal stresses. A combination of solidification shrinkage stresses and thermal contraction imposes strains on the semisolid network.[1–4] Several intrinsic factors affect solidification cracking. These include weld metal composition, freezing temperature range, solidification structure, amount and distribution of low melting phases, thermal and contraction stresses, and degree of restraint.[2–4] Various theories were proposed to understand the mechanism of hot tearing in casting based on stress, strain, and strain rate,[5–12] and some of these theories have been applied for hot cracking during welding. Many studies proposed that criteria based on strain and strain rate were more useful to explain hot tearing than stress-based criteria.[6–9] Strain-based model was first proposed by Pellini,[6] which was the basis for hot tearing criteria proposed by Clyne and Davies.[7] Currently, the strain rate is accepted as a critical factor[8] and is the basis of modern theories of hot cracking. The Rappaz–Drezet–Gremaud (RDG) criterion is the first two-phase model, considering fluid flow and the solid network deformation.[11] This physically based criterion offered a successful hot-cracking susceptibility (HCS) index to grade alloys.

Sindo Kou[13–15] simplified the RDG criterion considering the occurrences such as individual grain formation, their lateral growth, and liquid feeding along the grain boundaries. A simple indicator, namely solidification cracking susceptibility index (CSI), was proposed, $|dT/d(f_s^{1/2})|$ near $f_s^{1/2} = 1$, where T is temperature and f_s is the fraction solid in the semisolid region. CSI represents the steepness of the Scheil solidification curve (T *vs* f_s) evaluated towards the end of solidification. It depends on the chemical composition and thermodynamic parameters of the alloy system. Cracking is attributed to two observations: (a) the growth rate of adjacent grains is slow; thereby, the bonding is delayed, and (b) the obstruction in the liquid feeding due to longer grain boundary channels. This theory can be used to rank different alloys and welding consumables to comment on their weldability. CSI ignores solidification rate (v_T) for simplicity of analysis. The Niyama

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criterion uses this parameter as $v_T = \dot{T}/G$, where $\dot{T}$ is the cooling rate, and G is the thermal gradient.[3] During welding, the growth rate and the thermal gradient are coupled. These two variables ($\dot{T}$ and G) are expected to change the CSI *via* the microstructure. The local curvatures determine the micro-segregation and, thus, modify the local solidification path.

The existing models used for solidification cracking criteria[16] are subject to assumptions on the shape and arrangement of dendrites in the mushy zone, unidirectional variation of phase fractions, the directionality of dendrite shrinkage, and flow of melt in the interdendritic regions. The sensitivity of these criteria concerning various process variables was discussed by Eskin.[16] Although these assumptions are reasonable, the practical requirement of identifying a parameter window for welding an alloy without forming cracks still needs to be addressed. Hence, we propose a combinatorial experimental approach using a high-throughput experiment and computer simulations to address the above requirement.

Al-Cu-Li alloys are widely used in aerospace applications due to their high specific strength and superior mechanical properties at room and cryogenic temperatures.[17] However, the weldability of lithium-containing aluminum alloys is limited, and fusion welding becomes a challenge.[18] Aluminum alloy 2195 was chosen in this study due to its moderate weldability.[19]

In this study, solidification cracking was correlated with macroscopic parameters and viewed with a simulation-backed perspective to overcome the limiting assumptions of the theory and the complexity of a fusion welding process. The strategy adopted here intends to avoid a direct dependence on the welding conditions and identify crack-free regimes in the parametric space of macroscopic thermal parameters ($\dot{T}$ and G). The requirement of material and the number of experimental studies to identify safe welding parameters decreased significantly with a high-throughput experiment. A protocol combining the high-throughput experiment and computer simulations is suggested to identify parametric regimes for a crack-free weld in typical fusion welding processes. Relating these parameters to corresponding welding process conditions is a common inverse problem specific to the end application.

The effect of macroscopic parameters, mainly cooling rate ($\dot{T}$) and temperature gradient (G), on the solidification cracking is demonstrated in this study. Various parameters such as thermal contraction resulting in thermal stress and strain, strain rate, shrinkage, pressure drop, liquid feeding, etc., play a vital role in cracking. Thermal gradient and cooling rate are considered as the most important in relating the welding process with the local solidification microstructure. Hence, the study is limited to a two-dimensional parametric space defined by these two parameters.

An open-source C++ library (OpenFOAM) was used to create a solver to simulate the evolution of the thermal field during the spiral welding process. The welding heat source model parameters were calibrated using the experimental thermocouple measurements. Through these simulations, thermal gradients and cooling rates can be extracted at any spatial point within the domain in the relevant temperature range. The cooling rate ($\dot{T}$) was calculated between the liquidus and solidus temperature regime. Similarly, the thermal gradient (G) is the average value between the liquidus and solidus during the time interval. Physical locations along the spiral weld are mapped to the thermal gradient and cooling rate measured at those locations. By identifying the coordinates of the parametric space mapped to the physical locations of the cracks, the critical window for crack-free welding was identified.

Solidification microstructure at any location in a weld is determined by the local thermal gradient and cooling rate. Phase-field simulations account for local curvature effects on the interdendritic micro-segregation. These simulations are for the specific alloy under consideration, use alloy databases for thermodynamic data, and take the local cooling rate and thermal gradient as inputs. Theoretical solidification models are successful only to determine idealized composition profiles at the tip of a dendrite. Over the last two decades, phase-field simulations have been increasingly viewed as realistic in quantifying the micro-segregations taking local thermal conditions into account. Microstructure evolution was simulated using the phase-field technique using MICRESS® with a live connection to the appropriate thermo-kinetic databases from Thermo-Calc®. Phase-field simulation was done to establish the microstructure evolution at the two extreme conditions of solidification parameters. The phase-field simulation quantified the micro-segregation in the interdendritic region (as a function of $\dot{T}$ and G). The most segregated composition, thus, obtained is used as the last-to-solidify liquid. CALPHAD approach was used to calculate the Scheil's solidification curve (T *vs* f_s) for the interdendritic liquid.

This study proposes an integrated approach linking a high-throughput experiment, process simulations, and microstructure simulations to determine the window of processing conditions for a crack-free weld. The approach has been validated using linear welds.

II. EXPERIMENTAL DETAILS

A. *Material Details*

The chemical composition of the 2195 alloy used in this study is given in Table I. The workpiece was a circular plate of 200 mm diameter and 8 mm thickness. The surface oxide layer on the plate was removed by mechanical milling and acetone cleaning just before welding.

B. *Design of Experiment*

A high-throughput experimental setup shown in Figure 1 was assembled. The workpiece with a wedge-shaped copper backing plate was mounted on a table rotating with a constant angular speed. The copper backing plate was made with a radius of 200 mm, and the thickness was varied from 5 to 20 mm linearly across

the diameter. Initially, the tungsten electrode tip was kept at the center of the workpiece with the required stand-off distance. While welding, the workpiece was rotated with a fixed axis of rotation. The electrode tip was translated radially outwards from the center of the workpiece to the edge resulting in a weld along a spiral path. The spiral, thus, generated was modeled as an Archimedean spiral. The parametric equation of spiral, thus, generated is given as follows:

$$y = ut.\sin(\omega t), \quad [1a]$$

$$x = ut.\cos(\omega t), \quad [1b]$$

where u is the radial velocity of the torch tip and ω is the angular velocity of the rotating table.

The experiment was conducted in the following sequence: (1) initiation of workpiece rotation, (2) initiation of radial motion of the torch, and (3) arc was struck after the torch displaced radially by a reasonable distance. This offset is to avoid burn-through due to low initial net velocity (v_{net}) that results in localized heat input.

The strategy behind the spiral welding path in a circular plate was (a) to acquire varying resultant velocity (v_{net}) along the path resulting in a broad spectrum of cooling rate, (b) to subject the alloy to severe restraint conditions of circular welds (compared to linear welds), and (c) to obtain a wide range of process parameters per unit area. Equation [2] shows the resulting velocity at various points. The variation in heat input caused by the varying exposure time is evident from the equation:

$$v_{\text{net}} = \sqrt{(\omega r)^2 + u^2}, \quad [2]$$

where r is the radial distance from the center of the workpiece. A wedge-shaped copper backing plate was used to avoid conducting experiments with different plate geometries and different welding speeds (shown in Figure 1). This configuration resulted in a broad range of cooling rates.

The process parameters such as the velocity of the rotating table (ω) and linear torch velocity (u) should be chosen in a way that it yields a broad range of v_{net}. Thus, this experiment could qualify to be called high throughput. Moreover, a sufficient gap was maintained between the spiral arms to avoid overlapping thermal fields due to the proximity of welds and sufficient temperature drop after solidification across the adjacent arms. The strategies adopted to determine the welding process parameters are the following:

Table I. Chemical Composition of 2195 Aluminum Alloy

Element	Cu	Li	Mg	Ag	Zr	Al
Wt pct	3.9	1.0	0.65	0.3	0.19	bal.

1. The linear velocity (u) should be in the vicinity of $v_{\min}$ ($u \sim v_{\min}$), which is evident from Eq. [2], to obtain the lower bound of the desired velocity range at the beginning of the weld ($v_{\min}$ is the lower bound of the desired velocity range).
2. In the present experiment, the target was to have 3 to 4 circular arcs in the spiral, hence $\omega.t_n \geq 6\pi$. Here, t_n refers to total welding time.
3. The constraint on the diameter (D) of the workpiece is given by $u.t_n \geq 0.5D$.
4. To avoid the overlap and large temperature variation: $\frac{2\pi.u}{\omega} \geq m.w_{\text{p}}$. Here, w_{p} is the width of the weld pool and $m \geq 2$ for a sufficient gap between the adjacent spiral arms.

Based on the constraints mentioned above, autogenous welding was performed with the following process variables: welding current 250 A, voltage 20 V, torch linear travel speed (u) 40 mm/minute, the velocity of rotating table (ω) 12 rpm, electrode diameter 4.0 mm, electrode stand-off distance 2 mm, and argon shielding gas flow rate 16 liters/minute.

The heat input per unit length along the spiral path was calculated using the following equation:

$$E = VI/v_{\text{net}}, \quad [3]$$

where E is heat input per unit length, V arc voltage, I welding current, υ_{net}—resultant velocity at locations along the spiral weld.

Thermocouples were plunged in the workpiece at a depth of 3 mm from the bottom. The temperature was recorded at 10 Hz frequency using a computerized data acquisition system and Labview® software. The thermocouple data were used to benchmark the simulations.

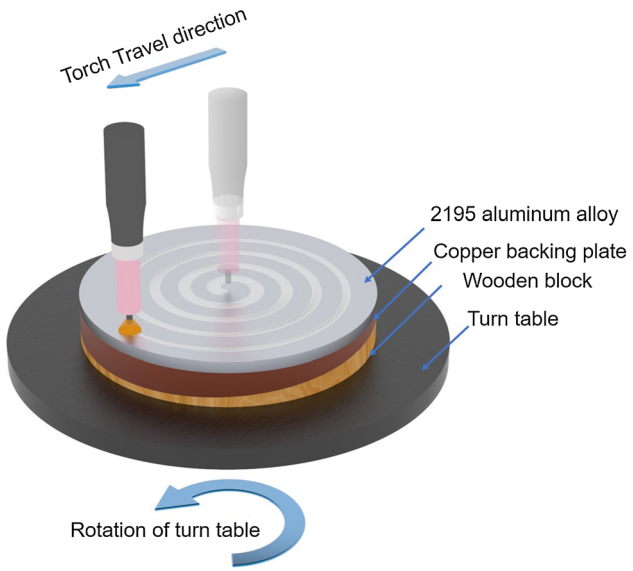


Fig. 1—Schematic of the high-throughput experimental setup.

C. *Metallography Analysis*

Weld specimens were sectioned, mounted, polished, and chemically etched using Keller's reagent for microstructural investigation. Fusion boundary and weld fusion zone were examined in the light microscope, Carl Zeiss Axio Lab A1®. Electron backscattered diffraction (EBSD) analysis was performed using the FEI Quanta 3D® FEG scanning electron microscope (SEM) to study the dendritic growth evolution at higher magnification. The scan area analyzed was 500 × 1200 μm at a step size of 0.4 μm. The elemental segregation at dendritic and interdendritic regions was investigated using an energy-dispersive X-ray spectroscopy (EDS) method using SEM, Tescan Vega 3®.

III. MODELING AND SIMULATION

A. *Welding Simulation*

Figure 2 shows the schematic illustration of the simulation domain and setup. The dimensions of the simulation domain are the same as the actual experiment. A full-scale simulation was performed to ensure that the experimental domain was captured completely, including the wedge-shaped backing plate and the entire spiral weld. The main assumptions of the heat transfer model were similar to those mentioned in an earlier study.[20] The governing equation for heat transfer accounting for heat loss due to phase changes can be written as follows:

$$\frac{dT}{dt} = \alpha\nabla^2 T + \frac{q}{\rho C_p} - \left(\frac{L_{hf}}{C_p}\right)\left(\frac{dF_l}{dt}\right) - \left(\frac{L_{hv}}{C_p}\right)\left(\frac{dF_v}{dt}\right). \quad [4]$$

Equation [4] is derived from Reference 21. Here C_p is specific heat capacity, ρ is the density of the workpiece, L_{hf} is the latent heat of fusion, L_{hv} is the latent heat of vaporization, α is thermal diffusivity, F_l is the liquid fraction in the mushy zone, F_v is vapor fraction, q is volumetric heat generation in the workpiece, T is temperature, and t is time. F_l is assumed to vary linearly with temperature, as suggested by Zhang *et al.*[21]

The volumetric heat source was based on the double-ellipsoidal model proposed by Goldak *et al.*[22] The heat source model is given as follows:

$$q_f = \frac{6\sqrt{3}Q\eta f_f}{abc_f\pi\sqrt{\pi}}\exp\left(-\frac{3x^2}{c_f^2} - \frac{3y^2}{a^2} - \frac{3z^2}{b^2}\right) \quad \text{if } z > z', \quad [5a]$$

$$q_r = \frac{6\sqrt{3}Q\eta f_r}{abc_r\pi\sqrt{\pi}}\exp\left(-\frac{3x^2}{c_r^2} - \frac{3y^2}{a^2} - \frac{3z^2}{b^2}\right) \quad \text{if } z < z', \quad [5b]$$

where Q is the rate of heat input, η is the efficiency of the welding parameter, and z' is the dynamic position of the heat source. The other parameters, such as a, b, c_f, and c_r were the front and rear ellipsoid semi-axis, respectively. The proportions of heat distribution among the front and rear ellipsoid were determined by values of f_f and f_r with a constraint given as $f_f + f_r = 2$. The condition $f_f/c_f = f_r/c_r$, as proposed by Nguyen *et al.*,[23] was imposed for continuity at z = z'. But here, since the weld path was a spiral, the front and rear ellipsoids depend on the angle in polar coordinates rather than the z coordinates.

The model incorporated only convective heat loss along the boundaries. The boundary condition given is as follows:

$$k\left(\frac{\partial T}{\partial n}\right) + h(T - T_o) = 0, \quad [6]$$

where k is the thermal conductivity of the system, h is the heat transfer coefficient, and T_o is the ambient temperature.

The simulation was carried out using a solver written in C++ using an open-source library, namely, OpenFOAM® (Open Field Operation And Manipulation) toolbox. The material parameters used for simulation are listed in Table II. The computational domain had ~10^5 finite volumes, and the time step used was 0.025 seconds. The grid size and timestep were sufficiently small to obtain an accurate temperature gradient and cooling rate, respectively. The values of the heat source parameter given in Eq. [5] were $a = 7$mm; $b = 2$mm; $c_f = 4$mm; $c_r = 8$mm; $f_f = 0.4$; $f_r = 1.4$; $\eta = 0.4$. The thermal profile measured by each thermocouple consists of three peaks corresponding to the proximity with the welding heat source traversing along the three arms of the spiral path. The duration between the three observed peaks, the relative heights of the three peaks, and the width of each of the peaks can be related to the welding heat source parameters. A set of simulations helped determine the trend of these characteristics of thermal profile with small changes to the heat source parameters. These numerical trends are then used as a guideline to bring the simulated temperature profile consistent with the experimentally measured thermal profile. The parameters, thus, obtained were again verified using experimental thermal profiles obtained by repeating the experiment under similar conditions.

After determining the heat source parameter and model verification, the temperature gradient and the cooling rate at various points across the spiral path were extracted. The cooling rate ($\dot{T}$) and temperature gradient (G) are calculated only towards the end of solidification at the respective location. The parameters were measured at a depth of 0.2 mm and at the center of the weld pool. The $\dot{T}$ was computed between the liquidus and solidus temperature regime. Similarly, the average value of G was obtained during the time interval between the liquidus and solidus. Since the thermal simulation was performed for the actual scale of the experiment, any change in the thermal gradient due to the accumulation of heat towards the end of the spiral welding was already accounted for.

B. *Microstructure Simulation*

Multi-phase-field modeling software MICRESS® was used to predict microstructure evolution and compositional gradients. This software was coupled with the

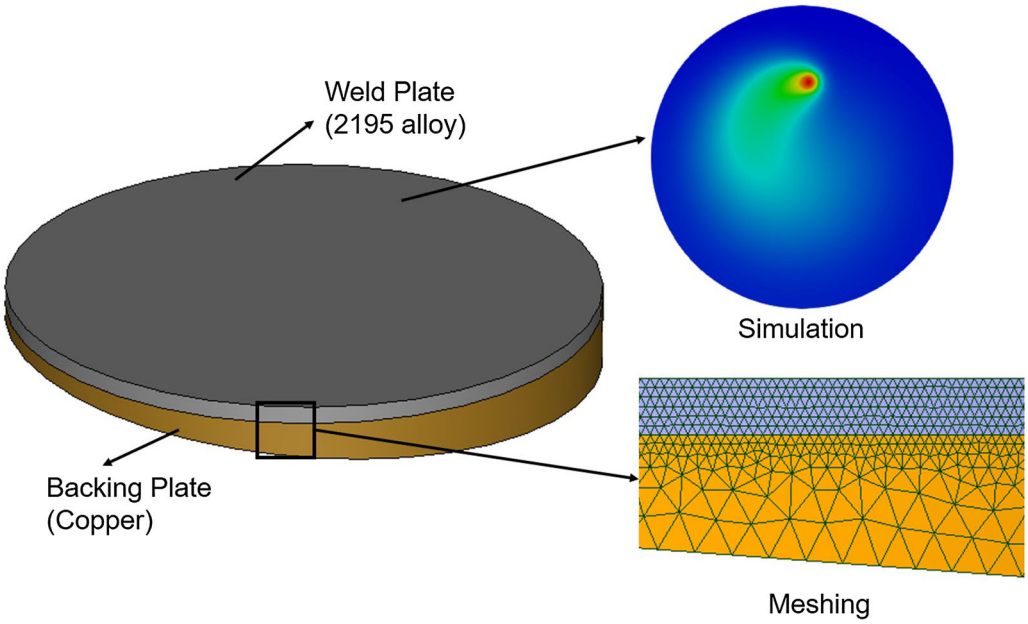


Fig. 2—Schematic representation of the simulation domain and setup.

Table II. Data Used for Heat Transfer Calculation[24,25]

Physical Parameters	Value
Solidus temperature, (K)	823
Liquidus temperature, (K)	933
Density (ρ), (Kg m^{-3})	3000
Specific heat capacity (C_p), (J Kg^{-1} K^{-1})	900
Latent heat of fusion (L_{hf}), (J Kg^{-1})	380000
Latent heat of evaporation (L_{hv}), (J Kg^{-1})	1050000
Thermal conductivity (k), ($Wm^{-1}K^{-1}$)	130
Thermal diffusivity (α), (m^2s^{-1})	4.81 x 10^{-5}
Heat transfer coefficient (h), (Wm^{-2})	5
Ambient temperature (T_o), (K)	303

Thermo-Calc® TCAL5 database to obtain equilibrium phases and the MOBAL4 mobility database to acquire diffusion coefficients of solute elements. Solidification parameters ($\dot{T}$ and G) obtained from the OpenFOAM® simulation were used as input parameters for the MICRESS® simulation. The complete set of equations, including the evolution of solute fields, solidification, diffusion, and anisotropy parameters considered in this study, can be found in the literature.[26,27] The parameters used in the MICRESS® simulation are given in Table III.

Regarding thermal conditions, the standard "frozen temperature approximation" with a one-dimensional thermal field was assumed. The problem case is the same as a typical directional solidification simulation where the thermal gradient and the cooling rate or growth rate are input parameters. The grid size was selected based on the convergence analysis, and the optimum diffuse interface width was selected based on the literature.[28] The simulation was performed on two-dimensional (2D) space with the domain size of 200 × 200 μm, and it was based on the moving frame following dendritic tips. The results are qualitative at this scale but preserve the trends in micro-segregation. These simulations provide quantitative estimates of micro-segregation in the inter-dendritic liquid during solidification under local thermal conditions. The most micro-segregated liquid alloy composition was picked manually from the output of the simulation, ensuring that the location lies within the interdendritic region and is representative of the last-to-solidify liquid. The solidification path of such a segregated liquid alloy directly determines the availability of liquid in avoiding hot cracking.

The simulation parameters used for the phase-field simulation are verified to avoid numerical artifacts and to reproduce the microstructure and the micro-segregation quantitatively and meaningful for the experimental results being analyzed. The diffusion coefficient values of Mg, Cu, and Li in Table III are direct results of Thermo-Calc®. The kinetic coefficient, interfacial energy, domain size, anisotropy of interfacial stiffness, and interfacial mobility values were optimized after systematic parametric study (not presented here) to get stable dendrite simulation.

IV. RESULTS AND DISCUSSION

A. *High-Throughput Experiment*

Figure 1 shows the high-throughput weld in a spiral path, where the weld starts near the center and proceeds outwards. Welding speed was controlled throughout to provide the gradual variation in the heat input. From Eq. [2], the exposure time reduces as the welding torch moves away from the center and reduces heat input. As the weld moved away from the center of the plate, width and depth were reduced but not monotonously due to

the wedge-shaped copper backing plate. Figure 3 shows the variation in heat input and weld depth as the torch moves away from the center of the plate.

Heat inputs measured at a radial distance of 25 and 85 mm from the center of the plate were 960 and 290 J/mm, respectively. Henceforth, these two regions at 25 mm and 85 mm are referred to as regions A and B, respectively, as shown in Figure 4(a). The weld depth was measured as 3.75 mm and 1.75 mm at regions A and B.

The solidification cracking was predominant at the initial portion of the spiral weld, as shown in Figure 4(b). The cracks were observed using a stereomicroscope from 25 mm to 48 mm (region A). Furthermore, fine micro-cracks were noticed through the stereomicroscope as the weld progressed. However, the density of micro-cracks reduced with the distance along the weld. No cracks were observed towards the end of the spiral weld (region B). The cracks are intergranular, seen along the interdendritic regions of the fusion zone (Figure 4(c)). Coniglio *et al.* reported that the increase in travel speed (low heat input) reduces the solidification cracking tendency. It is related to stress around the weld pool, grain structure, and mushy zone extent.[29] A similar trend was observed in our high-throughput experiment on 2195 alloy.

B. *Microstructural Characterization*

The velocity of liquidus isotherm (v_T) and temperature gradient (G) control the solidification microstructure. The velocity of isotherm (v_T) is related to the solidification growth rate (R) by the relation, $R = v_T$ (Cos α), where α is the angle between the direction of v_T and R at a particular point at the solid–liquid interface.[30] At the weld center, $R = v_T$ and $R = 0$ at the weld interface. The parameters G and R are, thus, dependent on the welding speed.

The G/R ratio decides the solidification mode, whereas G^*R determines the coarsening of the solidifying microstructure.[31,32] If the G/R ratio is high, there is less constitutional undercooling, resulting in planar or cellular growth. While the G/R ratio is low, constitutional undercooling is high, which results in columnar dendritic and equiaxed growth. The product G^*R indicates the cooling rate of solidifying weld. If the cooling rate is high, the time available for coarsening is limited, leading to a fine-scale microstructure. The cooling rate influences the distribution of alloying elements between dendritic and interdendritic regions. The size of the dendrite microstructure and cooling, thus, affect even phase formation in the solidified weld.

Specimens from the spiral welded plate were extracted from the cracked, and crack-free regions denoted as regions A and B, respectively. Figure 5(a) shows coarse columnar dendrites growing towards the weld center. Figure 5(b) reveals fine columnar dendrites. The average length of columnar grains at regions A and B is 235 and 99 μm, respectively. At region A, heat input was high due to a combination of slow welding speed and slow cooling rate. As a result, the G/R ratio at region A is high compared to region B. It resulted in coarse columnar dendrites that are more susceptible to solidification cracking. Conversely, in region B, the heat input was low; therefore, fine dendritic microstructure was produced, which is less susceptible to cracking.

The size and shape of the grains have a greater influence on crack susceptibility. Cracking in coarse columnar grains at region A was attributed to the following reasons[33–35]: (a) In coarser grains, the grain boundary area is lesser, so the low melting eutectics spread continuously along the grain boundaries and weaken the structure to resist cracking. (b) Contraction stresses produced during solidification and the strain

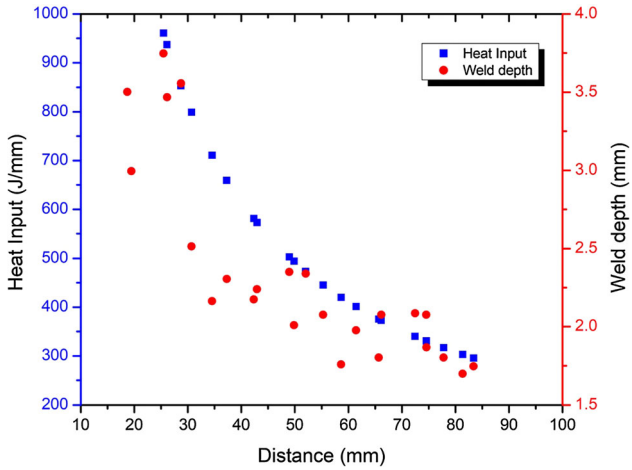


Fig. 3—Variation of heat input and weld depth with distance from the center of the workpiece. The heat input was calculated using Eq. [3].

Table III. Data Used for Microstructure Simulation

Parameters	Value
Diffusion of Mg in melt and dendrite (cm^2/s)	3.645×10^{-4},3.645×10^{-8}
Diffusion of Cu in melt and dendrite (cm^2/s)	$1.031x10^{-4}$, $1.031x10^{-8}$
Diffusion of Li in melt and dendrite (cm^2/s)	2.231×10^{-4}, 2.231×10^{-8}
Kinetic coefficient between melt and primary dendrite (μ, cm^4/Js)	4.000×10^{-2}
Interfacial energy between the melt and primary dendrite (σ, J/cm^2)	1.000×10^{-5}
Domain size, cell size (μm)	200×200, 0.25×0.25
Coefficient delta—Anisotropy of interfacial stiffness	0.3
Coefficient delta—Anisotropy of interfacial mobility	0.1

distribution are concentrated on the limited grain boundary area. In contrast, fine grains accommodate the stresses adequately and relax the stresses through the interdendritic regions. (c) Unlike coarse columnar grains, the fine grain structure slows down the initiation of interdendritic feeding.[36] (d) In the long-grain boundary channels of coarser grains, the permeability of the dendritic network becomes small, and the fluidity of the melt drops at the deep mushy zone. Therefore, feeding and healing of incipient cracks turn out to be difficult.[13,36] (e) Finer the grain size, the thickness of the liquid film in the interdendritic region reduces, which increases the capillary pressure required for crack propagation. When the capillary pressure lowers below a critical value, it generates cavities and initiates cracking. The thickness of the interdendritic region at region A was 0.63 to 0.87 μm and at region B, 0.33 to 0.49 μm.

In Figures 5(a) and (b), close to the fusion boundary, non-dendritic equiaxed grains were noticed ahead of columnar dendrites. Gutierrez *et al*. reported the formation of a similar structure in 2195 aluminum alloy welds. It was suggested that as adjacent regions of the fusion boundary and weld pool experience a temperature difference, the segregation of Al-Zr dispersions occurs towards the boundary. These dispersoids act as heterogeneous nucleating agents and produce non-dendritic equiaxed grains.[37,38]

The columnar dendrites observed near the fusion boundary are perpendicular to the fusion boundary. Figures 6(a) and (b) show no epitaxial growth from the fusion boundary. It could be because the formation of non-dendritic equiaxed grains through heterogeneous nucleation near the fusion line could restrict epitaxial growth. Random misorientations are observed between the base metal and fusion zone microstructure.

Weld centerline microstructure reveals equiaxed grains at regions A and B, as shown in Figures 7(a) and (b). At a specified welding speed (V), the growth rate (R) is maximum at the weld centerline and minimum at the fusion boundary. Similarly, the thermal gradient G is less at the centerline compared to the fusion line. As the cooling rate ($G.R$) is high and the solidification time is shorter, the grain structure is finer at the centerline than at the fusion boundary. A significant difference in grain size was observed between regions A and B.

Table IV shows elemental analysis at dendritic and interdendritic regions of A and B, which depicts a significant difference in solute segregation. The micro-segregation pattern is a time-dependent phenomenon primarily affected by the cooling rate and heat input. At region A, the solute (Cu and Mg) segregation was relatively less since the weld metal cools slowly (cooling rate is slow), and sufficient time was available for solute redistribution.

C. *Simulation*

The model verification was confirmed by comparing the thermal profile obtained from the experiments with simulations. Thermal profiles at different locations were obtained experimentally, and their corresponding simulated thermal profiles were in good agreement with each other. Figure 8(a) shows a typical thermal profile, and the corresponding position of the thermocouple in the welded plate is marked in Figure 9(a). The three peaks observed in Figure 8(a) correspond to positions when the heat source comes closer to the point. The curves flatten, and the peaks reduce when heat source parameters are increased to make the heat distribution relatively wider. There was good agreement between experimental data and simulation results. Hence, the

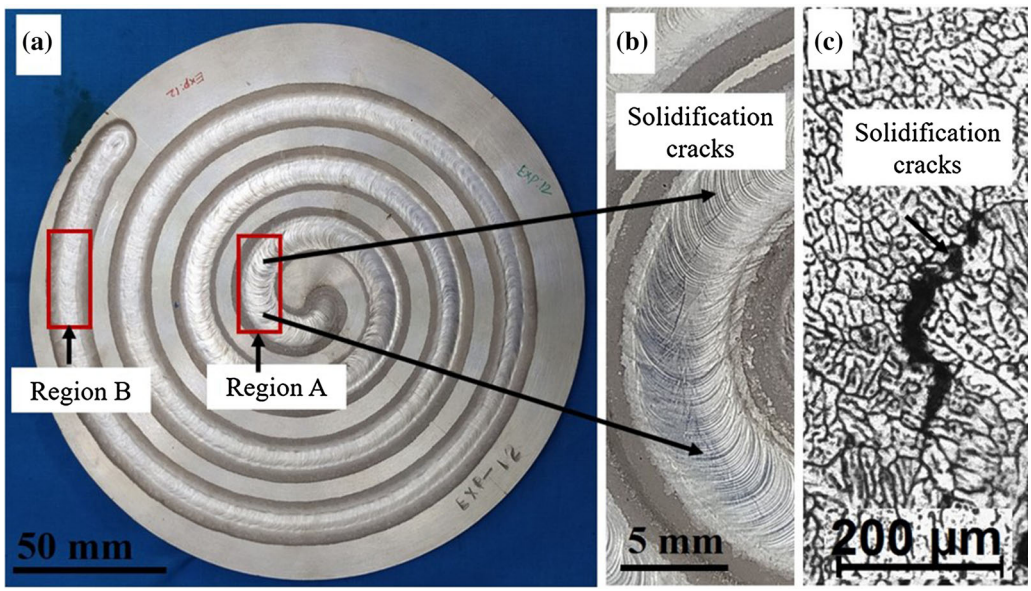


Fig. 4—(*a*) 2195 alloy spiral weld made by high-throughput experiment, regions A and B are cracks and no-crack regions, respectively; (*b*) Macroscopic view of solidification cracks; and (*c*) Optical microstructure of solidification cracks.

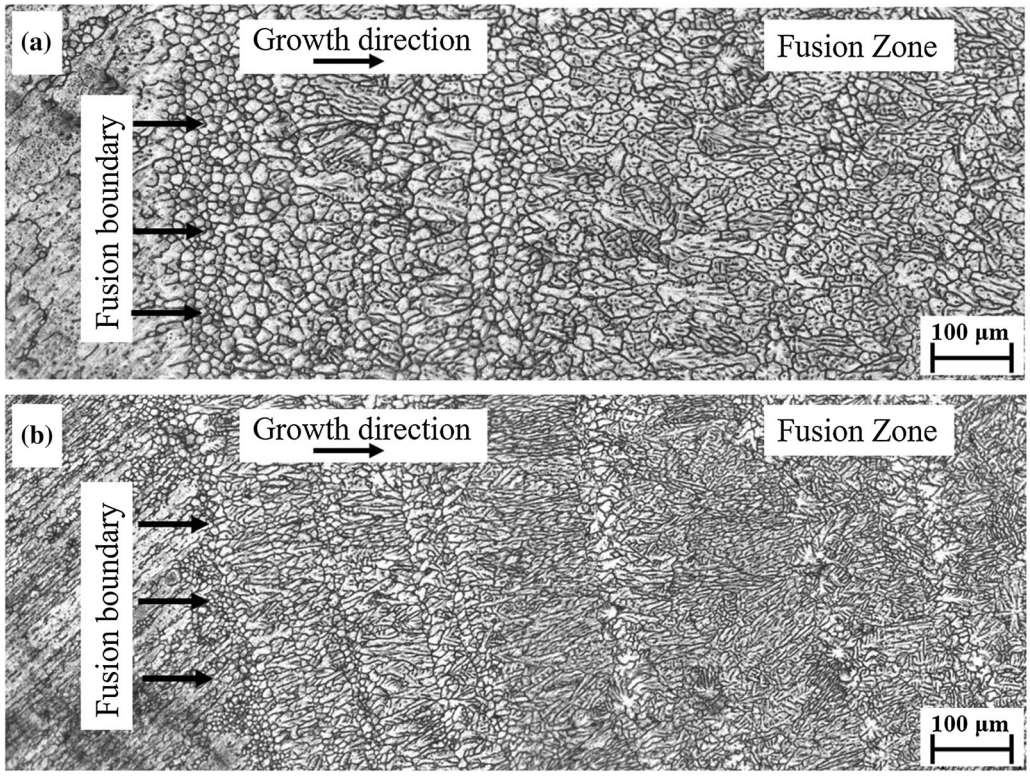


Fig. 5—Optical microstructure of fusion boundary and fusion zone: (*a*) region A; (*b*) region B.

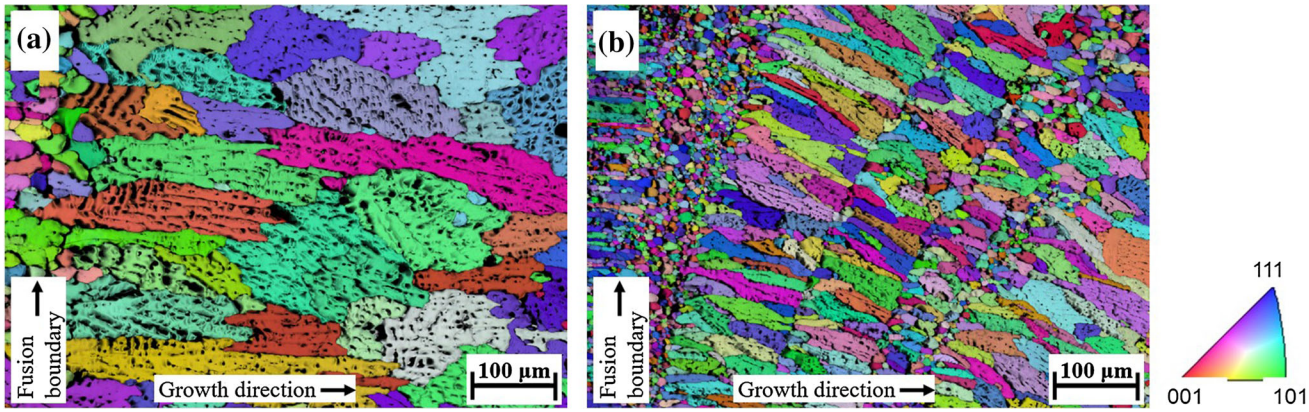


Fig. 6—Inverse pole figures (IPF) show dendritic growth evolution: (*a*) region A; (*b*) region B.

temperature gradient and the cooling rate calculated from the numerical simulations could be used to map locations along the spiral weld to a parametric space. Figure 8(b) shows the simulated thermal profile at the cracked location and the center of the spiral (overall arc length center). The cooling curve is steeper at the center compared to the cracked location, resulting in a comparatively lower cooling rate at the cracked location.

The two parameters were calculated for all the points shown in Figure 9(a). The coordinates of the cracked location obtained from the experiment are also highlighted in Figure 9(a). Note that these coordinates vary based on the experimental conditions. The parametric map with these two parameters was constructed, as shown in Figure 9(b). A broad spectrum of these two parameters can be observed in Figure 9(b); it supports that this experiment's claim to be a high-throughput one.

The critical parametric window (CPW) for cracking was identified by correlating the cracks observed in the experiments and mapping those locations over to the parametric space. Cracks were produced in the region where the cooling rate was approximately less than 110 K/sec (comparatively lower cooling rates), and the temperature gradient was between 1.2 and 2.2 $\times 10^4$ K/m. Cooling rates and the temperature gradients falling in CPW are susceptible to solidification cracking. Thus,

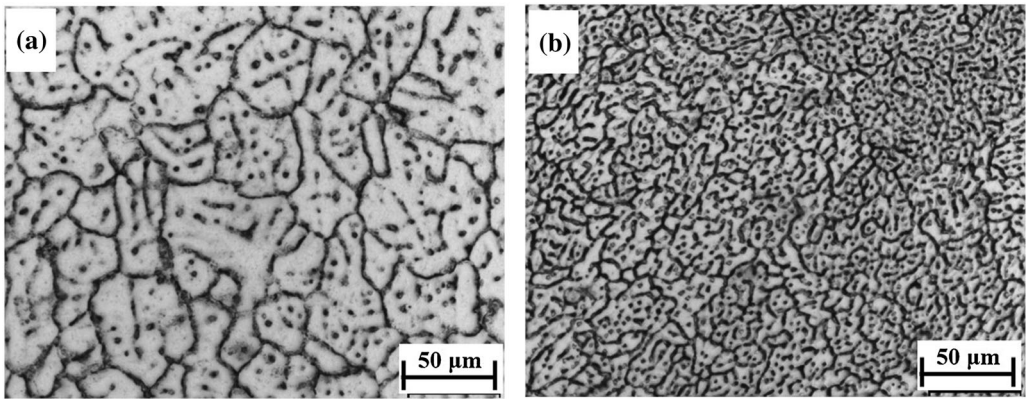


Fig. 7—Optical microstructure at weld centerline: (*a*) region A; (*b*) region B.

Table IV. Elemental Analysis in the Interdendritic and Dendritic Region of A and B

Element (Wt Pct)		Al	Cu	Mg
Region A	interdendritic	83.5	11.1	2.0
	dendrite core	98.7	1.0	0.3
Region B	interdendritic	77.4	16.4	2.6
	dendrite core	98.2	1.2	0.4

the CPW is independent of the actual process conditions and is a characteristic feature of the alloy solidification subject to the constraints of the welding process.

The cracking at the low cooling rate observed in Figure 9(b) is consistent with Eq. [7] given in[13]

$$V_{\text{local}} > \phi\sqrt{1-\beta}\frac{d\sqrt{f_s}}{dT}\frac{dT}{dt} + \phi\frac{d\left[v_z\left(1-\sqrt{1-\beta\sqrt{f_s}}\right)\right]_{x\to 1}}{dx}. \quad [7]$$

Here, β is solidification shrinkage, $v_z$is the liquid feeding velocity, f_s is the solid fraction, T is temperature, t is time, and ϕ is grain spacing. A low cooling rate makes the RHS smaller, and hence, V_{local} can easily exceed the threshold to cause cracking. The parameter ϕ depends on the thermal gradient and cooling rate *via* the solidification microstructure.

The regions where the two parameters ($\dot{T}$ and G) fall within the CPW can be identified for any welding process by carrying out the process simulations. Once the spatial regions susceptible to cracking are identified, ascertaining corresponding welding process parameters from the range in the parametric space of cooling rate and the thermal gradient is a standard inverse problem in heat transfer.

For validation, heat input at regions A and B in the spiral weld was adapted to produce linear welds. It was evident from Figures 10(a) and (b) that cracks were observed in linear welds having heat input of 960 J/mm, and no cracks were noticed at 290 J/mm. Thus, the parametric range obtained from spiral welding and the simulation were validated using linear welds.

Solidification parameters ($\dot{T}$ and G) influence the dendrite morphology, micro-segregation, and thus, the freezing range of the last-to-solidify liquid. Consequently, solidification cracking is affected by these parameters.[39,40] Microstructural simulation using the phase-field technique could bring out these correlations. The input parameters for solidification using MICRESS® simulation were obtained from macro-scale simulation of the welding process using OpenFOAM® simulation. $\dot{T}$ and G at region A were 80.43 K/s and 112.5 K/cm, respectively, whereas, at region B, the values were 189.5 K/s and 184.1 K/cm, respectively. Figure 11 shows the microstructure evolution during solidification in the two regions. Primary dendrites grow with the significant liquid channel between primary arms in region A and the imminent bridging of secondary arms in region B. The length of the liquid channel in the interdendritic regions was longer in the cracked regions than in the uncracked area. The length scale of the dendritic microstructure was finer in the regions that are crack free. The average primary dendritic arm spacing (PDAS) at regions A and B is 81 and 55 μm, respectively.

Figure 12 shows the solute concentration in the last-to-solidify liquid obtained from the simulation at regions A and B. The last-to-solidify liquid refers to the liquid present in the interdendritic region at the terminal stages of solidification. The micro-segregation of solute elements was lower in locations corresponding to the cracked region. Thus, the susceptibility is correlated with the composition of the last-to-solidify liquid. Cracks are observed in region A where the higher PDAS shows that the isotropic permeability of the mushy zone indicated by secondary arm spacing is not an appropriate indicator for cracking. However, during alloy solidification, there is a dynamic coupling between coarsening of the dendrite arms and natural flow induced by resultant shrinkage combined with Marangoni flow driven by local curvatures. This implies that simple analytical correlations based on permeability may not be sufficient to predict cracking.

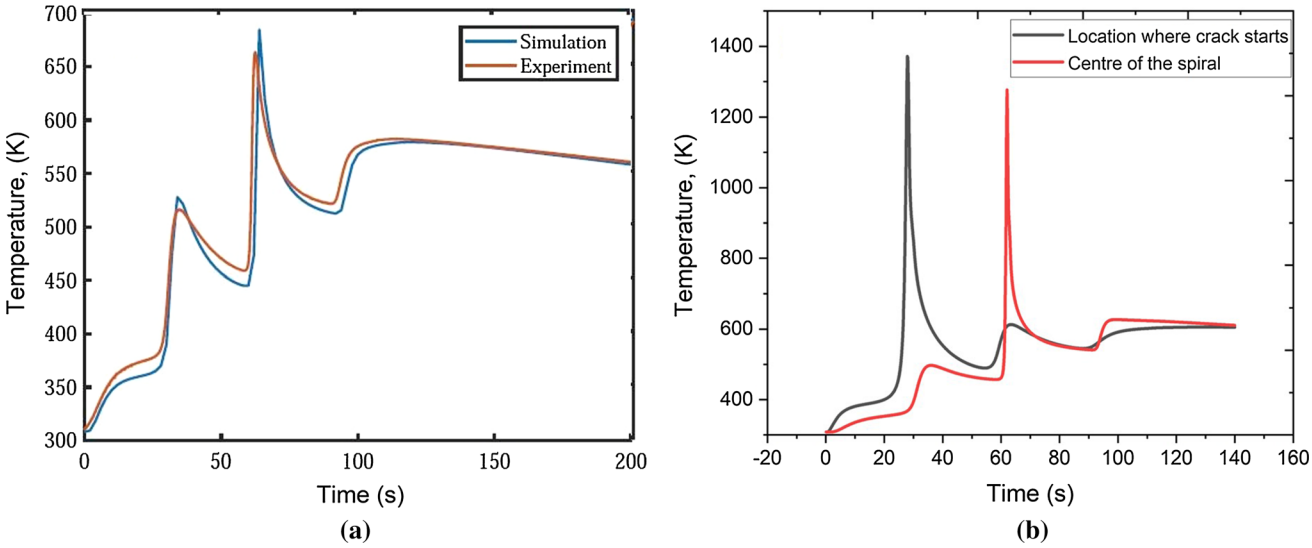


Fig. 8—(*a*) Comparison of experimental and simulation thermal profiles; (*b*) simulated thermal profile at the cracked location and the center of the spiral (overall arc length center).

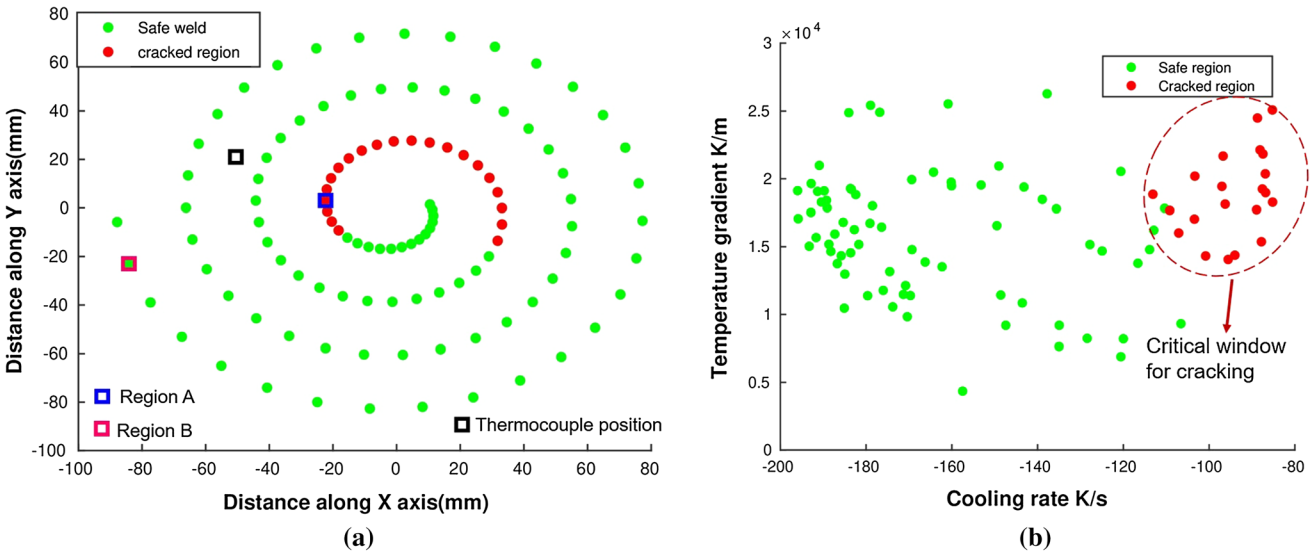


Fig. 9—(*a*) Physical mapping of points along the spiral weld path. The highlighted points belonged to the cracked region; (*b*) Parametric map of the temperature gradient (G) versus cooling rate for points shown in (a). The colored points are the mapping of the cracked region.

During solidification, the dendritic tip rejects the solute elements and increases the solute concentration in the liquid. Due to the slow cooling rate and lower thermal gradient at region A, a longer time is available for the solute to diffuse into the dendrites, whereas at region B, solute segregation occurs. The amount of micro-segregation and the composition of last-to-solidify liquid are also functions of curvature of the dendrite tip. In general, due to a faster cooling rate, dendrite tip velocity is high, and the curvature effect of the dendritic tip is stronger. The secondary arms tend to coalesce sooner with adjacent arms.[41] On the other hand, at slower cooling rates, the curvature effect is less pronounced, the secondary arms grow more rounded, and the coalescence of arms is delayed.

Figure 13 shows Scheil's solidification curve plotted using liquidus temperature as a function of the fraction of solid (f_s) for regions A and B, namely, the cracked and crack-free regions. The initial composition for these two curves is taken from the last-to-solidify liquid as calculated by the phase-field simulation using the local thermal gradient and cooling rate. The slope change at the terminal stage of the solidification curve corresponds to the interdendritic eutectic liquid. It was reported that during the non-equilibrium cooling of 2195 alloy, the eutectic compounds formed from the liquid are

Al_2CuLi, Al_2Cu, and Al_2CuMg.[42] The amount of these eutectic phases in the last-to-solidify liquid depends on the concentration of the last liquid to solidify. The higher the solute concentration, the amount of eutectic phase formation is more. At region A, the amount of interdendritic liquid is less than at region B. Also, the length of the liquid channel is longer in region A. The combination of a longer liquid channel and insufficient interdendritic liquid to feed the gap between dendrites leads to solidification cracking in region A. At region B, the length of the liquid channel is small, and the interdendritic liquid is available sufficiently over a larger temperature range to feed the channel; thereby, solidification cracking is prevented.[43] As per Kou's criterion,[13] the crack susceptibility index (CSI) is $|dT/d(f_s^{1/2})|$, which is the slope of the Scheil's solidification curve, near $f_s^{1/2} = 1$. The slope values for region A and B are 140 K and 125 K, respectively. These data corroborate Kou's criterion, where the higher slope value indicates the higher cracking tendency.

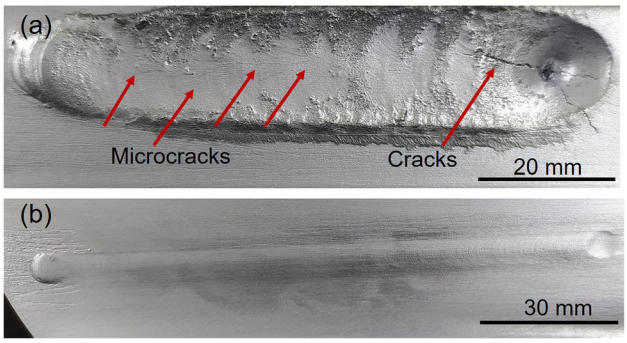


Fig. 10—Linear welds produced from extreme heat inputs of the spiral weld (*a*) 960 J/mm and (*b*) 290 J/mm.

The solute content of the liquid that freezes last is affected by the local microstructure, depending on the two parameters ($\dot{T}$ and G). Analytical criteria that do not consider these effects cannot suggest a parametric window for crack-free solidification. Hence, the solute composition in the last-to-solidify liquid taking microstructure as a function of cooling rate appears to be an important factor in correlating the crack susceptibility for a given alloy across a range of parameters during a welding process. An integrated approach combining high-throughput experiments, process simulations, and phase-field simulations of microstructure

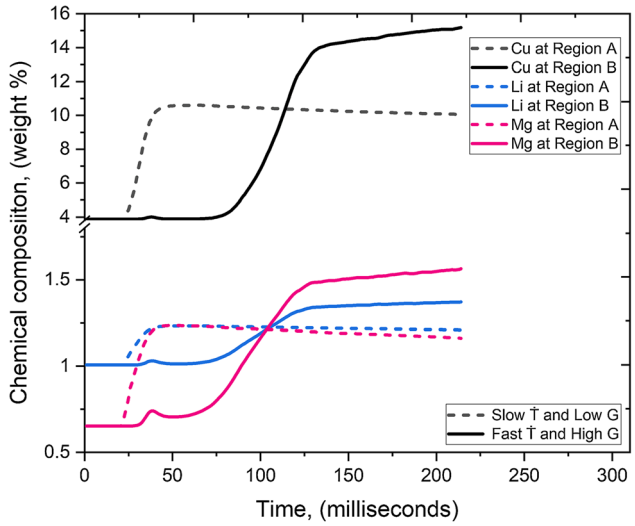


Fig. 12—The solute concentration in the last-to-solidify liquid obtained, from the phase-field simulation at regions A and B. The composition measured is an average value at the center of the interdendritic region at a constant time step.

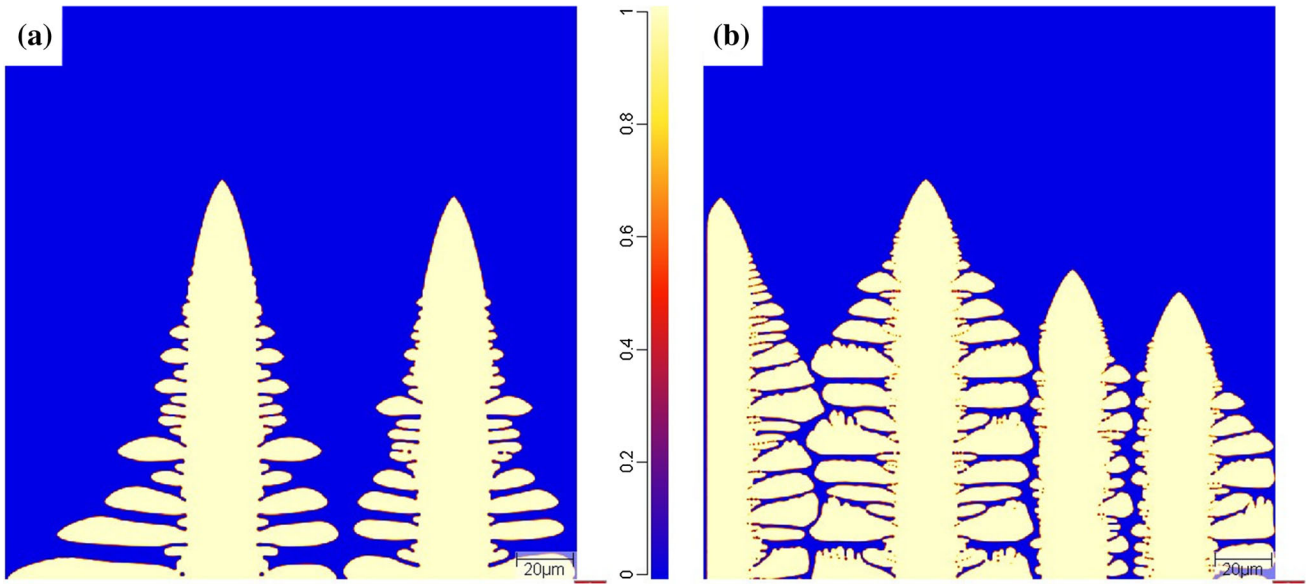


Fig. 11—Simulation results showing the microstructure at (*a*) region A; (*b*) region B. The color map shows the variation of the phase-field order parameter Φ, (Φ = 0, for liquid and Φ = 1, for solid) (Color figure online).

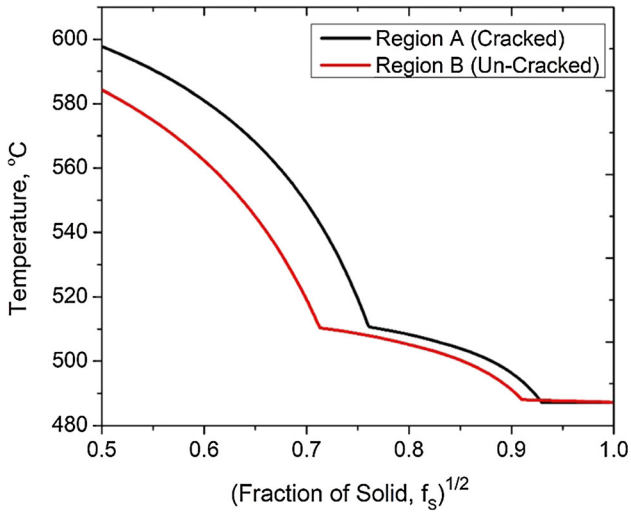


Fig. 13—Scheil's solidification curve—(Temperature versus fraction of solid, $f_s^{1/2}$) for region A and region B.

could help address the gap in understanding hot-cracking susceptibility.

V. CONCLUSIONS

1. A high-throughput experiment was proposed to study cooling rate ($\dot{T}$) and thermal gradient (G) on solidification cracking. A wide range of $\dot{T}$ and G was obtained from the parametric map simulation and mapped over to the locations along the spiral weld. Furthermore, it was found that cracking was predominant at the high heat input region of 960 J/mm.
2. Cracked regions were characterized with coarse columnar grains where the length of the liquid channel (interdendritic regions) was observed to be longer than the uncracked region. Due to the slow cooling rate in this region, more solute redistribution and less micro-segregation occurred. The trends in microstructure evolution and solute distribution predicted by phase-field simulation were confirmed in experimental characterization.
3. The composition of the last-to-solidify liquid for a local microstructure depends on the parameters ($\dot{T}$ and G) and is a useful indicator for crack formation. In region A, exposed to high heat input, the length of the interdendritic liquid channel was more, and the fraction of eutectic constituent was less. Therefore, the insufficient liquid could not fill the interdendritic channels and resist cracking.
4. For alloys graded as crack-prone by the criteria currently accepted in the open literature, there is a need to predict a window of processing conditions for crack-free welding. This work demonstrated that a high-throughput experiment combined with simulations could evolve a quick procedure to determine the safe window of the parameters ($\dot{T}$ and G) for crack-free welding. Further, this approach can be extended to other engineering alloys that are graded to have limited weldability.

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CONFLICT OF INTEREST

The corresponding author states that there is no conflict of interest on behalf of all authors.

REFERENCES

1. W. Messler Jr.: *Principles of Welding*, Wiley, New York, 2007, pp. 373–453.
2. S. Kou: *Welding Metallurgy*, Wiley, New York, 2003, pp. 263–300.
3. B. Chalmers: *Principles of Solidification*, Wiley, New York, 1967.
4. J.C. Lippold: *Weld. Metall. Weldability*, 2014.
5. T. Zacharia: *Weld. J.*, 1994, pp. 164–72.
6. Pellini W.S: *Foundry*, 1952, vol. 80, pp. 125–33.
7. T.W. Clyne and G.J. Davies: The Metals Society, London, 1979, pp. 275–78.
8. N.N. Prokhorov: *Russ. Cast. Prod.*, 1962, vol. 2, pp. 172–75.
9. K. Nakata and F. Matsuda: *Trans. JWRI.*, 1995, vol. 24(1), pp. 83–94.
10. N.N. Prokhorov and N.N. Prokhorov: *J. Japan Weld. Soc.*, 1971, pp. 109–17.
11. M. Rappaz, J.M. Drezet, and M. Gremaud: *Metall. Mater. Trans. A.*, 1999, vol. 30A, pp. 449–55.
12. N. Coniglio and C.E. Cross: *Metall. Mater. Trans. A.*, 2009, vol. 40A, pp. 2718–28.
13. S. Kou: *Acta Mater*.https://doi.org/10.1016/j.actamat.2015.01.034.
14. J. Liu and S. Kou: *Acta Mater*. https://doi.org/10.1016/j.actamat.2016.03.030.
15. J. Liu and S. Kou: *Acta Mater*. https://doi.org/10.1016/j.actamat.2016.12.028.
16. D.G. Eskin and L. Katgerman: *Metall. Mater. Trans. A.*, 2007, vol. 38A, pp. 1511–19.
17. R.J. Rioja and J. Liu: *Metall. Mater. Trans. A Phys. Metall. Mater. Sci.*, 2012, vol. 43, pp. 3325–37.
18. A. Kostrivas and J.C. Lippold: *Int. Mater. Rev.*, 1999, vol. 44, pp. 217–37.
19. M. Agilan, G. Phanikumar, and D. Sivakumar: *Trans. Indian Inst. Met.*, 2018, vol. 71, pp. 2667–70.
20. M. Sheikhi, F. Malek Ghaini, and H. Assadi: *Acta Mater*. https://doi.org/10.1016/j.actamat.2014.09.002.
21. W. Zhang, G.G. Roy, J.W. Elmer, and T. DebRoy: *J. Appl. Phys.*, 2003, vol. 93, pp. 3022–33.
22. J. Goldak, A. Chakravarti, and M. Bibby: *Metall. Trans. B.*, 1984, vol. 15B, pp. 299–305.
23. N. Nguyen, O. A, M. K, S. N, and M. Y: *Weld. J.*, 1999, vol. August, pp. 265s-274s.
24. *ASM Handbook - Properties and Selection: Nonferrous Alloys and Special-Purpose Materials*, Volume 2., ASM International, 1990.
25. E.A. Starke: in *Aluminum-Lithium Alloys: Processing, Properties, and Applications*, N. Eswara Prasad, A.A. Gokhale, and R.J.H. Wanhill, eds., Butterworth-Heinemann, 2013, pp. 3–26.

26. M.R. Rahul and G. Phanikumar: *J. Alloys Compd.*, 2020, vol. 815, p. 152334.
27. I. Steinbach: *Model. Simul. Mater. Sci. Eng.* https://doi.org/10.1088/0965-0393/17/7/073001.
28. J. Eiken: *IOP Conf. Ser. Mater. Sci. Eng.* https://doi.org/10.1088/1757-899X/33/1/012105.
29. N. Coniglio and C.E. Cross: *Int. J. Adv. Manuf. Technol.*, 2020, vol. 107, pp. 5025–38.
30. P. Schempp, C.E. Cross, A. Pittner, G. Oder, R.S. Neumann, H. Rooch, I. Dörfel, W. Österle, and M. Rethmeier: *Weld. J.*, 2014, vol. 93, pp. 53–59.
31. S.A. David and J.M. Vitek: *Int. Mater. Rev.*, 1989, vol. 34, pp. 213–45.
32. S.A. David, S.S. Babu, and J.M. Vitek: *JOM.*, 2003, vol. 55, pp. 14–20.
33. N. Coniglio and C.E. Cross: *Int. Mater. Rev.*, 2013, vol. 58, pp. 375–97.
34. Dickhaus C.H, L. Ohm, and S. Engler: *Trans. Am. Foundry Soc.*, 1994, pp. 677–84.
35. M. Braccini, C.L. Martin, M. Suéry, and Y. Bréchet: Matériaux Tech., https://doi.org/10.1051/mattech/200088050019.
36. Z. Tang and F. Vollertsen: *Weld. World.*, 2014, vol. 58, pp. 355–66.
37. A.E. Gutierrez and J.C. Lippold: *Weld. J.*, 1998, vol. 77, p. 123.
38. A.D. Kostrivas and J.C. Lippold: *JOM.*, 2004, vol. 56, pp. 65–72.
39. L. Wang, N. Wang, and N. Provatas: *Acta Mater.*, 2017, vol. 126, pp. 302–12.
40. L. Wang and K. Wang: *J. Manuf. Process.*, 2019, vol. 48, pp. 191–98.
41. S. Sandhya, R. Mahemaa, and G. Phanikumar: Procedia. *Mater. Sci.*, 2014, vol. 5, pp. 408–15.
42. A. Hekmat-Ardakan, E.M. Elgallad, F. Ajersch, and X.-G. Chen: *Mater. Sci. Eng. A.*, 2012, vol. 558, pp. 76–81.
43. P. Jiang, S. Geng, X. Shao, G. Mi, C. Wang, H. Wu, C. Han, and S. Gao: *JOM.*, 2019, vol. 71, pp. 3223–29.

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